ActInf Textbook Group ~ Cohort 3 ~ Meeting 23 (Chapter 10, part 2)

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Duration: 1:09:31 | **Uploaded**: Unknown **Transcript method**: auto_caption

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all right greetings everyone thanks for joining it is August 15th and we're in our final discussion recorded discussion for cohort three next week we'll have uh kind of unrecorded discussion also this will be the day of the Symposium so very auspicious day to have been concluded upon but today we're going to talk about 10 and probably anything else so Ali and Neil want to leave with anything or and or we can go to ask Mel's question first well I've quickly put in the last three questions but we can I think we can get through those so you know we can start with a smiles all right I'll just read it um and then I happen to hear anyone's thoughts or or more details now since I believe in Occam's razor I denied the meaning of awareness my suggestion is accessible cognition and inaccessible cognition it looks like we have an operating system and implicit memory that's not available such as the procedural memory and emotional memory I have some doubts about the rest of the words like Consciousness and awareness and this is about the question the opening quotation of the chapter beginning at the end in general we are at least aware of what our minds do best Neil or Ali but she's on my my last question which was what is sentient about active inference uh that's it's put in the chapter title uh what I've suggested as a teaser the chapters really just summarizing uh it's it doesn't really say anything about Consciousness I don't think and what Carl is it all seems to be getting more involved in discussions of Consciousness but I think he always maintains a distance between that and works with with people who are concerned with that but it's it's more something where active inference is is a framework for um well to help people who are looking at sentence and consciousness uh but doesn't really say anything specifically about it and so it seems to me awesome Ollie wanted anything on that well yeah as far as I know sentience um inactive inference literature is used uh in rather technical sense in which um sentient beings are just [Music] they're the beings that are responsive uh to the external stimuli so that's as simple as it gets uh and uh because of its Simplicity it encompasses a whole array of things as being sentient and uh it's not necessarily uh restricted to biological sentience or uh I don't know uh any thing akin to to the level of human intelligence or human sentience and so on so it can basically be applied to include artificial sentience and everything even uh virtual sentience such as I don't know kind of thing that's just been implemented algorithmically the software and stuff so I get as it it was a subject of the rather heated debate a few months ago uh I mean the meaning of the sentience is used in active influence literature but in all honesty it's not a novel use either because uh this the same very definition of a sentence that's used in active inference is almost exactly the same as one of the definitions from Oxford English dictionaries so uh it's not as controversial as it seems but about the other terms such as awareness and Consciousness well uh I don't believe we have any sort of consensus around those words yet in active inference literature although in recent papers around mum or Mom model of active inference um Maxwell and others are trying to converge a consensus on the word on the notion of Consciousness in terms of active inference yeah what grade topics and also how cool with chapter 10 published about a year and a half ago maybe written in the previous months leading up to one and a half years ago how how there's certain similarities and continuations of the discussion yeah Neil I really like the the point about having us talk how how active inference how it stays it's unopiniated to the extent that it's composable as a describing system those properties are kind of like convex concave and there's a lot to say probably about why certain expectations are around certain kinds of X of accounts scientific accounts or just certain kinds of rhetorical accounts like why that why some are some way and some are not that's that's this active influence of social sciences um but that's the whole thing having the descriptive framework cartography compositional cognitive cartography as per the category Theory work and and that whole area especially coming into play and the mechanics and the Dynamics it's like Newtonian mechanics whatever you say about it within the system it doesn't have a preference for something moving left or right or up or down or something like that that's just a bizarre feature to bake into a measurement framework so when we look at active really is presented in this textbook whether as a behavioral researcher like the figure 9.2 metabasian so whether we approach it from this kind of like classical behavioral researcher or just researchers perspective or we approach it from the constructive perspective with uh chapter six and all of that from six or from nine uh science and engineering eventually maybe it'll come to be seen as an account of the cartography so hence that that Divergence with how the the book lands of course this isn't apologetics it's not to say that there's like a perfect text or anything like that but I think that's kind of the area I know that's a bunch of random thoughts but it's chapter 10 these are the funny things about the book that we only really come into relation with when we really stay in the game and and compare how it actually was so we can look at some specific parts of with the kinds of perspectives that that people mobilize today based upon like for example only in the last few days different experimental studies I'll bring in one link just as an example uh of course only only in for educational use this LinkedIn post friston's AI law is proven fep explains how neurons learned anonymized to protect the innocent um Kristen's or everyone's slash no ones AI law proven is explains how what and if you think about this in light of of how we've been thinking about 10 then and now how does this new artifact from today seem to you yes Ali well actually I haven't read the article in its entirety but uh from what I've seen uh it's not fundamentally different from the previous work done in establishing a kind of empirical basis for active inference but I'm not sure why it gained so much attention now because uh to my mind it's just a continuation of the previous work done by the same group of authors and establishing these kinds of empirical bases for active inference but I can understand that it can seem as a kind of culmination of this trajectory of this line of research but uh I'm I'm not sure if there's something fundamentally new or I don't know unprecedented explored in this recent paper or it's uh it's a kind of um I mean as I said a culmination of all the previous efforts done uh in this area but on the other hand uh of course because it's published in nature uh it would attract lots of media attention and as we we've seen in the past few days on Twitter or I don't know X or anything uh it's the kind of uh you know I I I'm not saying it's kind of um uh it's reached the level of hyperbole or something but uh again I'm not sure if it's quite beneficiary for active inference literature if these kinds of I mean this kind of news gets out of proportion and delivers I mean it it wouldn't be able to deliver as much as it promises uh so as a case in point that Denise holds uh image here seems to me a little bit out of proportion I don't know what you think but maybe it's just my personal opinion that it's always better to be uh to to keep some restraint about these things and that let let everything loose because it has it can have dire consequences for any future research if uh it it it doesn't get what it promises or it doesn't deliver what it promises or something like that so that's just my personal opinion on that excellent I'll leave very measured very diplomatic Neil yeah I'm I'm only just looking at the article now yeah I don't think that I'll just bring in one other um cultural artifact that that like uh back in the game I just remember being like again everything everyone's Bay is optimal so it's it's I I'm not saying I would have done anything differently I'm not saying anything like that I just want to point to these um historical um image Meme and and everything this was the year that the Alias interview with Martin forger and Carl had been um well all you need is the title but what else a funny thing I just thought okay I know that it's the article is not ever defined by the headline Etc genius neuroscientist so okay speculative but singular and then the generality connected to the requirements to understand a specific mind which I think is the same cognitive associativity between like Nate that's a whole Marco blanket LaPlace approximation the Tony mechanics like that we always talk about so that's my first take is just like okay well yeah he may work here or there but um also Stanford doesn't say that it's Gordon's law of collective Behavior but also there's a lot of different Norms around communication and another adjacency to kind of explore to based upon how we're talking about active like to say the least as at least instrumental and realist but also as I again all of you have brought up in previous discussions like there's so many other uh perspectives interpretation so if a statistical paper came out some gene expression study from 1992 or some uh height blood pressure risk related study Pearson's progression law is proven linear regression explains how something happens that's also playing into this kind of like scientific positivism again not necessarily the place for the whole falsificationism of science disproves doesn't prove but to to see here's the um what the science do how is science related to law whose ideas are whoms what acronyms do you already have to know an account of a natural process legitimizing um or endorsing even applications of some similar technical method in a different setting a lot of Rich image Meme and and uh media fun ahead of us yeah Andrew throwing memes the LK 99 I don't even know what if uh someone's like this is a uh you know room temperature super cognitive process actually LK 99 was exactly what I had in mind when I talked about um yeah promising something that can't be delivered yeah again there's so much more to say on this like it's just the complex consequences of different regimes of attention it's so so much more complex and like good or bad or for whom or over what time scale and uh surfing the 100 foot wave with popularization and also connecting people like just to be explicit to come to the textbook group and to volunteer be an intern at The Institute then there's an actionable path from going from like being enthralled by a podcast to working in that tradition it's all happening in the ecosystem and we're just seeing how we see it from the a few of us who are here just in this slash any given moment all right Neil which one of your questions do you think is most relevant well perhaps perhaps the most controversial one is the how important is the theory of free energy minimization so we could could do that oh yeah really this is a question of qualitative versus quantitative the notion of free energy is very quantitative but is it how essential is it it's a way of formalizing and Computing something so so useful from an engineering perspective isn't it just enough to say well the brain is it's uh a load of neurons in the brains are working in a a self-organizing fashion with uh some I want to use fireman's phrase or ratchet and pool mechanism to to always be want to be moving gradually towards a better solution so then we have this this uh quantity free energy that's that's incomputable effectively uh that's that we can say well that's that's a macroscopic um quantity but it's it's it doesn't get used anything in any way um as as it does in some machine learning algorithms it's uh so you know so it's purely epiphenomenal it doesn't it doesn't actually do anything other than well to help understand what's going on um so how how important is yeah that whole look uh quantitative side of things to the to all of the uh the qualitative aspects of that active inference so implicit is is my my thought that it isn't uh it isn't essential yeah this is a very interesting question it's a very JF claudger type point like how different can it be than outside in variational Bays contrasted with jf's constructed Lego symbolic implementation with no no equations no variational updating so I think that's a really important area to explore Ali go for it I will actually looking at active inference framework um I mean superficially we can we can see lots of similarities and even um I don't know probably some uh identical uh conceptual components uh with some other Frameworks such as predictive processing and uh predictive coding and so on but in my opinion the main difference of active inference uh one of the main difference of active influence from the other related Frameworks is exactly what fep enables active inference to do namely it puts the optimization of surprises as the main character of the whole scenario so you see this is uh the novel perspective of active inference as compared to all the other Frameworks because uh of course in all the other perspectives we have the Notions of I don't know anticipating actions and so on and basing some decision making to basing the the decision making on the prediction and anticipate the anticipation of actions and also perception but uh I mean the way fep reframes this whole scenario and as I said puts the surprisal optimization uh to the Forefront as the main character uh makes it much more elegant on one hand and on the other uh it makes it much more broader than all the other Frameworks because we know that predictive coding predictive processing they're mainly restricted to biological intelligence or biological I don't know cognition and whatever but uh we know from uh at least 2019 that the same mathematical technology can be applied to a whole range a whole array of um ontological entities and uh it's not fundamentally restricted to uh biological sentience or intelligence so this is the key reframing and the key novel perspective that fep provides to active imprints and if if you look at it I mean these kinds of fundamental reframings uh and uh somehow pivoting the whole structure of a scientific theory based on just a single reframing of a of an essential component is something that uh that can be witnessed throughout the whole history of science time and time again and some of the most profound insights of that's been gained in throughout the entire history of modern science uh has been they've been gained from this exact mechanism of changing the perspective uh based on some redefinition and some reconceptualization of the established facts and established theories so in that sense active inference and its um underlying fep uh to my opinion can be seen as I mean it's comparable to all of these quote-unquote revolutionary scientific theories uh in that regard probably thank you great chapter 10ism super Super Bowl set I think this helps differentiate us again while we're in chapter 10 which is essentially doing this exact work wrapping up and connecting non-technically without figures just connecting the dots 10.3 it also connects us to the cybernetics side it's non-contentious that cybernetic systems can be modeled input output it's also non-contentious to do multi-agent simulation and then coming from whether just a pure pragmatic perspective in a finite world or increasingly from the holographic approach and the Chris Fields physics is information processing physics of boundaries information processing and thermodynamics and boundaries also kind of converging on a multi-agent thermo informational input output framework and when we're talking pedagogically sometimes it could be like well it's like input output but you could have like just an oscillator like just a sine wave or something like that now also I'm sure the sine wave can be understood as doing the path lease action all of this that is the generality of the framework and it really does pivot or have a fulcrum as you described Ali as the actual censoring of surprisal in principle in literally in principle and in fact slash practice slash deed detractable bounded probably approximately correct bounded rationality in this situation variational Bayesian mechanics but Quantum Hardware software could be used surely other kinds of computational implementations can be used but the censoring of surprisable in principle with the bounded approximation of surprisal with all of the work of the bounded surprisal being bounded surprisable about a given model can't talk like generally about the L2 Norm for some cohort of linear regressions you haven't named yet that's just not enough information to to say further so the theory takes an essential normative Focus on surprisal minimization and therefore bridging to the practice with uh attracted by uh attractable optimizable bound no contention with other cybernetics Frameworks they want to explain observables as a loss function that's awesome perceptual control theory RL they want to have structures or unstructured or fixed or dynamic cognitive architectures also all good um they want to include action and sense making and use a variational autoencoder or a variational Bayesian modeling imperative specifically that's interesting Gray Zone question is that implicit fep if it's literally already providing a particular partitioned example that would be very interesting dark fep literature search but it's actually those Works which arguably are used every day in terms of statistics algorithms by reframing the scenario it's like hey we don't have to reduce uncertainty slash bound uncertainty slash maximize model evidence on the reward function turn the surprisal bounding on itself purely statistically that doesn't mean it's going to be perfect because first there's the information computation bounds Chris Fields uh Wolfram all of that and then just practically your implementation is going to be like just realistically put together by you and however you actually put it together sorry talking about in principle like exploring blueprints for buildings discussing very broadly maybe even usefully over broad possible buildings in principle without even wondering if they can be built or anything like that or are we really in the game talking about motor disorders in this context of this situation very very different Waters fep and octymph though it can be uh assured as far as we believe that there is a fundamental relationship that seemingly we learn more about brings in more pieces because it's a pretty common useful as well as converged upon Forest utility already enabling a lot of the shorter term work to be scholarly and of course all pending Etc et cetera Etc but that's as Ali has pointed out as they call it the structure of scientific revolutions and so in that way that's an interesting and provocative uh area to explore Ollie um also on a related point and uh getting back to the previous discussion we had about the nature of Consciousness sentence awareness and so on uh so uh Philip ball has this recent book uh called The Book of Minds which is a 500 page tone but he recently gave a talk to the Royal Institution um it's titled uh let me see yeah it's what is a mind uh so in this talk he he concentrate on just one chapter of this whole book uh namely uh the space of sorry yeah the space of possible Minds so more often than not we think of all of these uh concepts related to mind such as intelligence agency sentience and so on as a kind of one-dimensional spectrum one-dimensional Continuum but uh Philip Paul here shows that uh it would probably be more um I mean conducive to further uh development the development of further future theories if we conceive a kind of multi-dimensional space of possible Minds which can have the access such as agency experience intelligence and Consciousness and it it draws a kind of multi-dimensional space such as this and I was thinking active inference can actually provide sufficient tools to Traverse this whole Space of possible Minds if one tries to put all of these different axes related to one of the key trade-offs that's been that unfolds through the central equations of active inference so basically uh I was thinking that probably or perhaps the central equations of active inference can be used uh as a kind of uh mathematical tool to Traverse this whole possible uh by using uh all the all the trade-offs that we've seen throughout the whole book uh such as explore exploit phenomena or other kinds of dilemmas that we've seen before so yeah I just wanted to share and this connection with you because I believe there can be some potential uh interesting connections between these two theories that's great another kind of recent uh connection I was just listening to a little of this machine learning Street talk with Stephen Wolfram and he starts talking about yes the rouliad and and all the other work and it's really also impressive how much he condenses it and how much they progressed and then he's like talking about a physics of the Observer and a framework for observation and I was just like that's where they need to kind of like Direct Time stamps clear ontologically structured rhetoric I mean we're in the farmer's market face still overall and how far away is it really from processing and analyzing on one on one branch processing the analyzing speech to develop appropriate paths and wayfinding and guiding and all even if it's Stephen Wolfram giving him the most informative on-ramps just like we would for anyone on the computational side and also to Neil's Point focus with what's the qualitative side and the true cognitive the phenomenological side the social side and I think that's one reason why um with with scholarly work like that of Mel Andrews and and Mal and others integrates discussion around model building it also draws upon the pragmatic and the abductive and the semiotic and a lot of other strongly process oriented we talk about process ontologies um all these different features being woven in at the very least compatibly but compatibly which is to say open-endedly articulated in principle that's of course we're just having discussions who knows what's over under confidence just just saying how it strikes me at this moment we wouldn't really want a better imperative based upon what we understand at least what I personally understand something along those lines that sets the stage for multi-perspectival modeling of diverse intelligences setting that composability it's like if it doesn't have composability and the unopinionated uh kind of nature collab as the collaborative project that's what we want collaborative project essential project critical infrastructure specific project Ryan Smith used a stomach uh electrode to test this hypothesis about the vagus nerve and the role of this like neuropeptide and this muscle uh pattern then we're we're in the standard scientific expanded scientific space in the process Theory Space chapter six chapter nine did you look at the whole system did you consider systematic factors what what trade-offs did you make is modeling is the process and pipeline reproducible do we have the code like in chapter nine can you make a figure about the experiment just like connected to the figures in the textbook like reviewer type functions but as we often bring up reviewers are not going into whether L2 Norm or L1 Norm was the statistically appropriate choice I know in ethology many times we just said we looked at this other paper or just this is a commonly used approach for this statistical technique putting a lot of weight and focus on the Last Mile and the specifics which is obvious like Pearson's not brought out every time somebody does or doesn't use an r squared to mobilize an arguments or make a decision that's so many levels away Einstein is not brought out except ingest when the planes are late a lot of a lot of um morphological and Behavioral development but one of the key practices um a one thing that Wolfram was saying was like there are certain things about this or that that have to be the case like he was talking about our continuous time perception even though it's the hypergraph like the discretized hypergraph at the lower level uh with that kind of like averages over all the computations I believe um something like that um and then like we're orders magnitude larger and slower we average out we core screen talked about that then the theory of the Observer and he's like yeah there might be like other things about communication or about observation or systems that also like must be the case so I think there's kind of like in terms of the anthropic principle slash anthropic fallacy it's almost like I believe that we will through chapter and Beyond like hey how many careers like this is not just hot takes on a podcast these are many many careers in many many directions of many many academic departments a lot of research programs startups a lot of funding Cycles around what is or isn't plausible AI winter um all the implications causes and consequences of cybernetics research it's like this is the umbrella that is on the table connected only to other accounts so barring things that we learn on a week by week basis so at least scale the pragmatic value to how relevant it is and then calibrate your epistemic scalar to what and how you do and don't know like we all do and don't know and only speak from our own experience we can have a cognitive modeling analysis what happens when somebody who is not to pick on any field or any topic or anything like that someone who's like oh I love that citation or like that's my job or like I know that really well or my dissertation was in that what happens in the moment of contextualization not to get too meta but again whether it's artisanal Farmers Market status and just like well anecdotally people who are um doing psychiatric work or who are interested in like for example chronic pain as like several of our very committed colleagues are like they prefer this that can be approached by kind of naively or expertly qualitatively and or quantitatively so we're in in the end of this um cohort of uh three of the textbook group so what is it really about the textbook how do we really want to think about its role in the ecosystem like impact on people's learning production of different materials for different consumptions and publishings what any of you think about that just to give one other random take just kind of unstructured thinking about it is like I was thinking of walking down the aisle at a bookstore and there's different Frameworks caching or vying for attention and I know it sounds obvious that it has to be differentiated but pointing out that there's differentiation on multiple axes as a consequence it sounds wild or weird or however you want to see it at first but the minimization of surprise has a central phenomena and that simple starting positionality or stance along with some other core memes that I think needs to be worked out and understood the relationship to like embody and all this other stuff that rolls downhill differently helps you rethink differently see a lot of phenomena maybe theoretical transdisciplinary synthesis is not your um trajectory in life that's totally chill obvious it doesn't need to be said but just to be clear that this is in that the just the the where and the when and the how and The Who in relationship to if the um actual which we saw uh at least the the movie poster for earlier in our discussion uh and also one of the points is uh um you see given the rapid development of active inference literature since the publication of this book uh I believe um it might not be a terribly bad idea if uh beginning from the next cohort uh we somehow begin to incorporate more uh actively some of the uh findings of recent literature into the discussions um I mean with this pace of development that active inference is experiencing right now uh it might sound a bit grandiose but I think it's not unlike uh the situation that quantum physics experienced uh in the early 20th century so uh as we know from the history of quantum physics uh almost every year and even every few months radically new uh understanding of quantum theory uh appeared which fundamentally shaped what we know as quantum physics and I believe active imprints kind of uh taking a similar trajectory because with of course not all of the um papers that's been published since the publication of this book but some of the uh groundbreaking papers that really shifted our understanding of what active inference is all about and uh somehow refined uh the more vague Concepts that we still see in this book and try to make them more precise and make it more coherent with the rest of the theory so uh yeah as I said in my opinion it might not be a bad idea to look at some of these developments from the next beginning from the next cohort thanks a lot Ali Neil and grows with any new area it grows it gets beyond the point where where any individual can keep track of things I was wondering I mean maybe the active inference Institute is effectively doing this already but I'm just imagining if if there was an annual report let's say The Institute uh produced an annual report providing an overview of of what happened uh the significant developments uh in in the last year so that you were talking earlier about you know there's so many different ways in that it touches on uh that people might well have some involvement to do with active inference but then you know this this tiny area uh how do you get them to have some bigger perspective whether there's education such as this book but how do you enable them to carry on maintaining that because because they can't be reading thousands of Journal papers or a year just on the active inference when that might actually only be a relatively small part of their their wider research whereas The Institute should be able to provide some oath is it it's not I'm not just thinking that a review of um something so said this and so said that but some sort of bigger perspective on on where on what's Happening so like a whole load of new art new Publications were in in this direction say social sciences and that to to help people thanks these are such good comments I really agree Ali and I and now actually kind of like pseudo-templated a little bit like a research Roundup so it's a seed project that's just one possible mechanism not to not jump to a solution or anything like that but like I also totally agree like yeah we fixed some Errata and typos we aggregated some people's input that's that's local refinement and it's it's adding value and so on but then okay now Jonathan and Ali have PDFs on this equation and that equation that's like and then you have sanjeev's work as all of you no The annotation work is infinite and I think we've in some ways laid out what certain paths look like like for example using the ontology seriously annotating equations with the verbal descriptions getting all the figures boxes tables eventually even paragraphs into tabular form like we have the massively collaborative possibilities and that's not to say they're even being underused at all by human and or non-human agents it's just part of a co-creation so it's not about just like um just like single dimensional perspectives on Consciousness or anything like that are limited if if not worse um multiple projects should speak to the bigger perspective I think there's the important observational role like here's how surprise was used in this paper at this time I associate that with the kind of trust finder architecture like rhetorical assertion not to get too technical but different expressions are made in the ecosystem ingressed and or intentionally we have different strategies and approaches and and projects that are anticipatory and maintenance oriented and historical to improve our sense making there are some interns who are doing such amazing work and that's only the fraction some fraction of people um who are not just in their own way and in their own game which is everyone so like what we're what we can really do I'm happy to have these fundamental conversations I I just I we're not going to be able to enumerate or evaluate or anything like that but par some you know keeping the par flame alive maybe with a discussion changing the time Ollie and I will have completed our background videos and then we can kind of assess like here's the rules here's the first versions of the background materials go through it at these you know small groups so there could be a lot of decentralized ways to continue that or at least just low preparation ways for different facilitators maybe we need cohorts who have Chris fields um maybe we need a special integration units and projects like that just having a reading group follow AVL in social sciences and follow Chris in in the you're going to be intersecting and and uh engaging and still on the fractal wave and we're always going to be only using the systems that we have can't be waiting for Future tools but we will use them as they arrive and just supporting the sustainability of that for the ecosystem is our another thought just as we're in kind of last minutes of 10. so it's like all this theoretical scope yeah it could just be another um node in the same space but the claim is kind of like it generalizes a lot of them relates a lot of them formally um maybe even envelops all of them at least socio socio-technically if not even way way way deeper and broader and and everything like that so it's like however much Theory scope you think this covers maybe you're super familiar with literature you're for every single citation you're like oh yeah you know exactly why it's there then this argument is poised to your adjacency because this is the perspective of the authors or if you're not super familiar with these areas it may be sufficient just to understand them as being like broad topics of Interest if not addressing importance systems in the real world but when we really think about these as being the major threads of accounts and statistics for cognitive behavioral systems it's not like there's some other like secret 11th Theory to just being like censored in the citation literature I mean I'd leave it to a meta-analysis to say more but but nobody reads this and says wow you're you're ignoring like this entire massive thing yeah maybe go into more detail on adaptive resonance Theory or go into more detail on this on one given area but when the broad threads are represented and the arguments are made formally in text based upon an implicit first the scope and the relevance of the learning has to be this theoretical scope and the consequences on systems of Interest doesn't mean it has to be overwhelming in the moment but people bring up real things in the textbook group and it is about real things because education is about real things I would like to hear any of your all's thoughts on chapter 10 now as we close don't worry authors will get the final word not us any thoughts foreign so many great comments and I know it's only the tip of the iceberg of what was like said or maybe it was never even said and then of course what said it's only the iceberg with with what is thought but it's so funny it was right back there with the sentience with the 11th round of the sentience debate and it's like is it over promising but it's a it's a different system of interest it's not the same thing as learning like learning epistemic value that's why it's sometimes fun just to learn about things for their epistemic value there's parts that we know has pragmatic value there's parts that we know have epistemic value what other imperative are you acting and inferring under and then one route that Ali highlighted very um particularly earlier was the central character main character energy for surprisal and it kind of uh close in that Nexus is the qualitative but but formal potentially qualitative compositional about existence existentials and that ties it to homeostasis and all ties it to observation and all of that it opens up another degree of freedom the Axel constants Mobius strip it's not about it's about how you got like this describes systems with persistence do you have a better descriptor but that doesn't mean that constructing it is a guarantee of future persistence you know past returns does not guarantee future returns or whatever they say there you may be able to say certain things but the existentialism is a natural fit in information landscapes because it kind of blurs it doesn't actually blur but it can but it conceptually integrates the measurement and the existential Anaconda common footing super General analytical representations the exact same kind from like variational physics and mechanics information Theory Quantum category Theory Dynamic systems all their notation special case all the math that people are using to describe systems of Interest compositionality recognition of new underlying special case phenomena just as Buckminster Fuller always pointed to recognition of special case phenomena upon generalization also related to the revolution model that Ali mentioned and that is the thing if you just have a particular partition but then you have them fitting an L2 Norm which is not unlike what a structural equation model is actually in structural equation modeling is used in SPM and I'm sure there's all kinds of different variants and stuff but it's like that's like a kind of massively linear parametric can include time series just like all the sarima methods and all of that so a lot of the same ability to describe like changing causal relationships little-known fact existed in SPM so a lot of the practice was totally totally addressed by SPM textbook various modules became increasingly recognized centrality of action the pragmatic turn in Neuroscience all that those uh recognitions uh were either articulations that were not understood to have already existed like the idea that you could just extend the kind of Coleman filter like predictive processing algorithm arising in video compression and then just like pop action in and get active inference with all of these even immediately apparent philosophical implications for people who are studying those topics and the maintenance of the key feature with the physics-based first principles singular imperative so recognitions of articulations that that um suggested also profitable and fruitful directions importance critiques and and re-understandings and developments uh not just related to notation for example generative model and generative process which as we've discussed more recently is not the way that it's often being discussed so does that lead to an entire secondary discussion or do we just continue forward mostly with such and such but the particular partition and the over the last years the refinement of so many different aspects such that we're basically like preparing to scramble not really how to even synthesize hence the ultra descriptive deflationary ontology at assertion based architecture that we hope is going to bring in a lot of diverse ways of knowing and remodeling and sense making there will be a lot of difference angles and overlaps like what if there's a statistics Professor who's just feels a certain way about this how does that situate them in relationship to the theoretical apparatus as understood by any other given person or by any other group of other people so how to really retain the singular Focus amidst all the specifics it's comforting to expect and prefer that we have the framework that is on the path towards that composability extensibility etc etc all the phenomena that we discuss and what's the other aspect to it and the textbook's not life so how to emphasize that too and really integrate it in people's living and learning make it what it can be so we could definitely think a lot and no specific dates for cohorts are set but maybe you know who knows social sciences cohort 2 Chris Fields cohort II par it's people's attention slash nestmate level contributions that actually embody those things and so after the acclimation of like the kind of whoa there's still so much work and that will continue to be it's been a total honor to Neil Ali and esmail and everybody else who contributed to totally to this cohort um any any last comments on cohort 3. fascinating Journey as always I mean the summary well summary chapter will always do that and sort of throwing everything there is in front of you but yeah there is so much to the whole to the subject isn't there I've brought in from so many different places it really is yeah we've all learned a lot it's been a very great end so I guess next week the the day of the symposium we can discuss more but we'll see what we may want to do or how we want to proceed in the coming um cohorts so thanks again see y'all next time bye thanks everyone thanks bye thanks